

8086 System Connections and BUS Timing

Course Teacher:

Md. Obaidur Rahman, Ph.D.

Professor

Department of Computer Science and Engineering (CSE)
Dhaka University of Engineering & Technology (DUET), Gazipur.

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Course Title: Microprocessors and Assembly Language
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Lecture References:

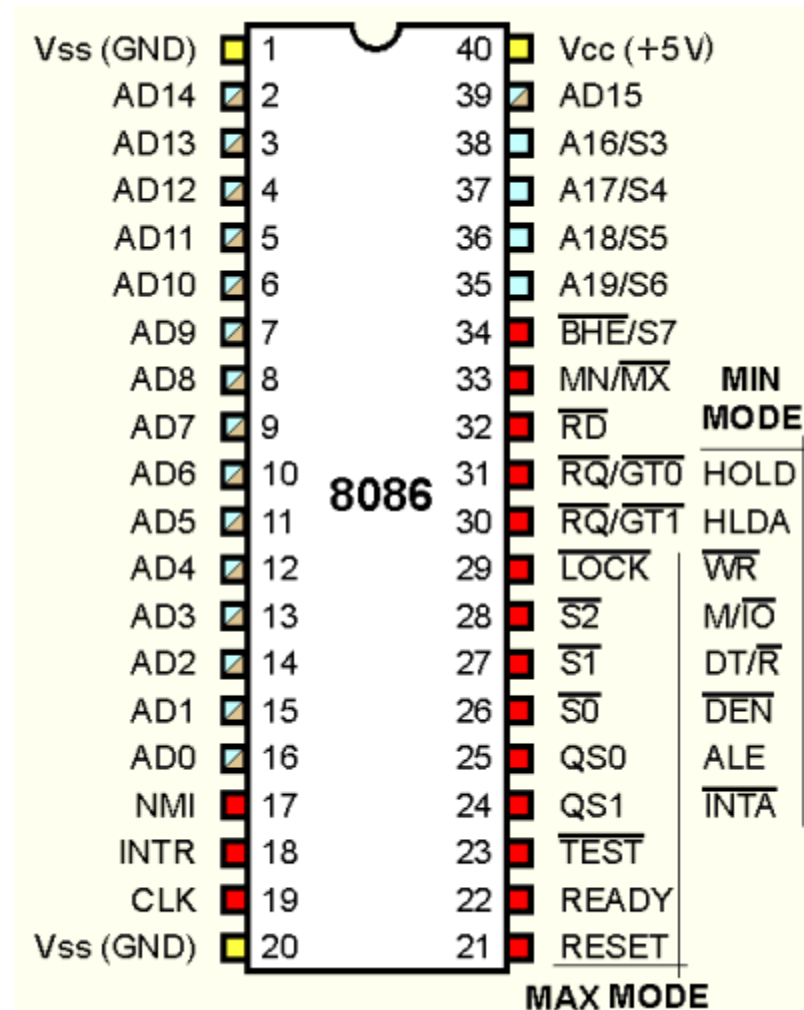
- ▶ **Book:**

- ▶ *Microprocessors and Interfacing: Programming and Hardware, Chapter # 7, **Author:** Douglas V. Hall.*

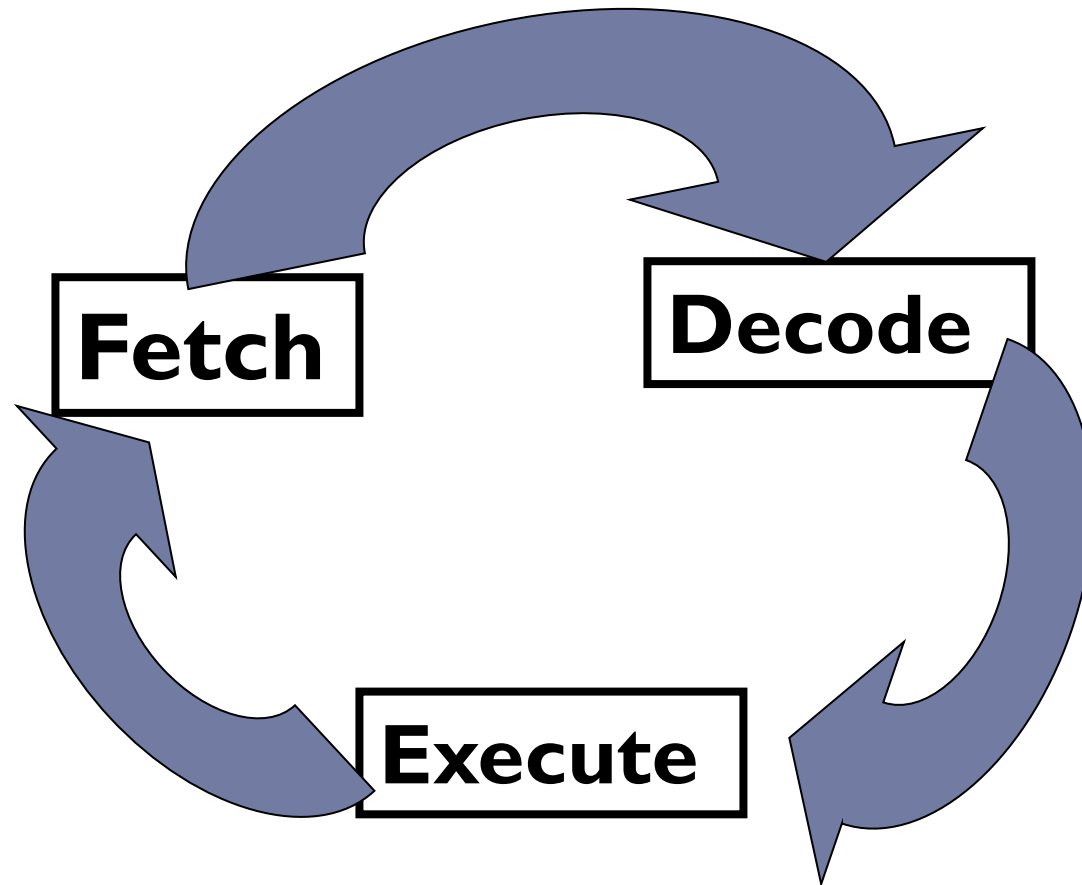
- ▶ **Lecture Materials:**

- ▶ M.A. Sattar, Microprocessor and Microcontroller, BUET.
 - ▶ Md. Omar Faruque, BRAC University.

Remember !! 8086 Pin Specifications



Microprocessor Operation

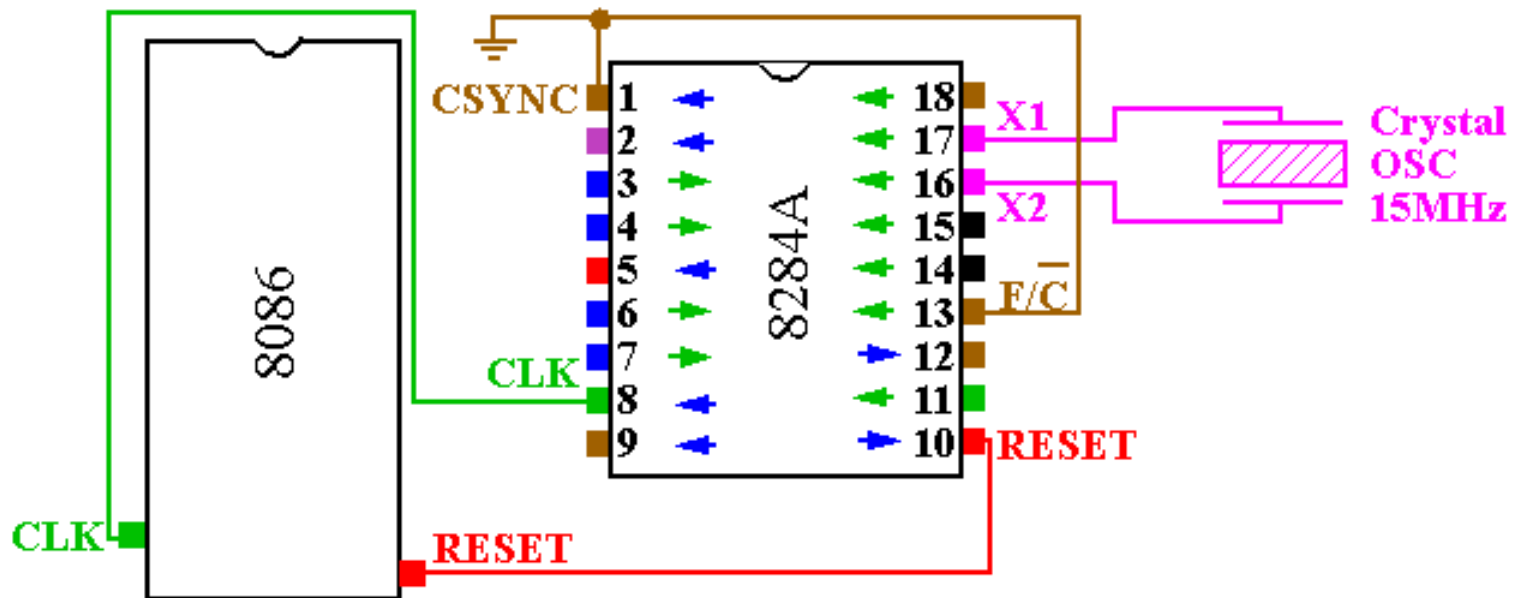


Microprocessor Operation

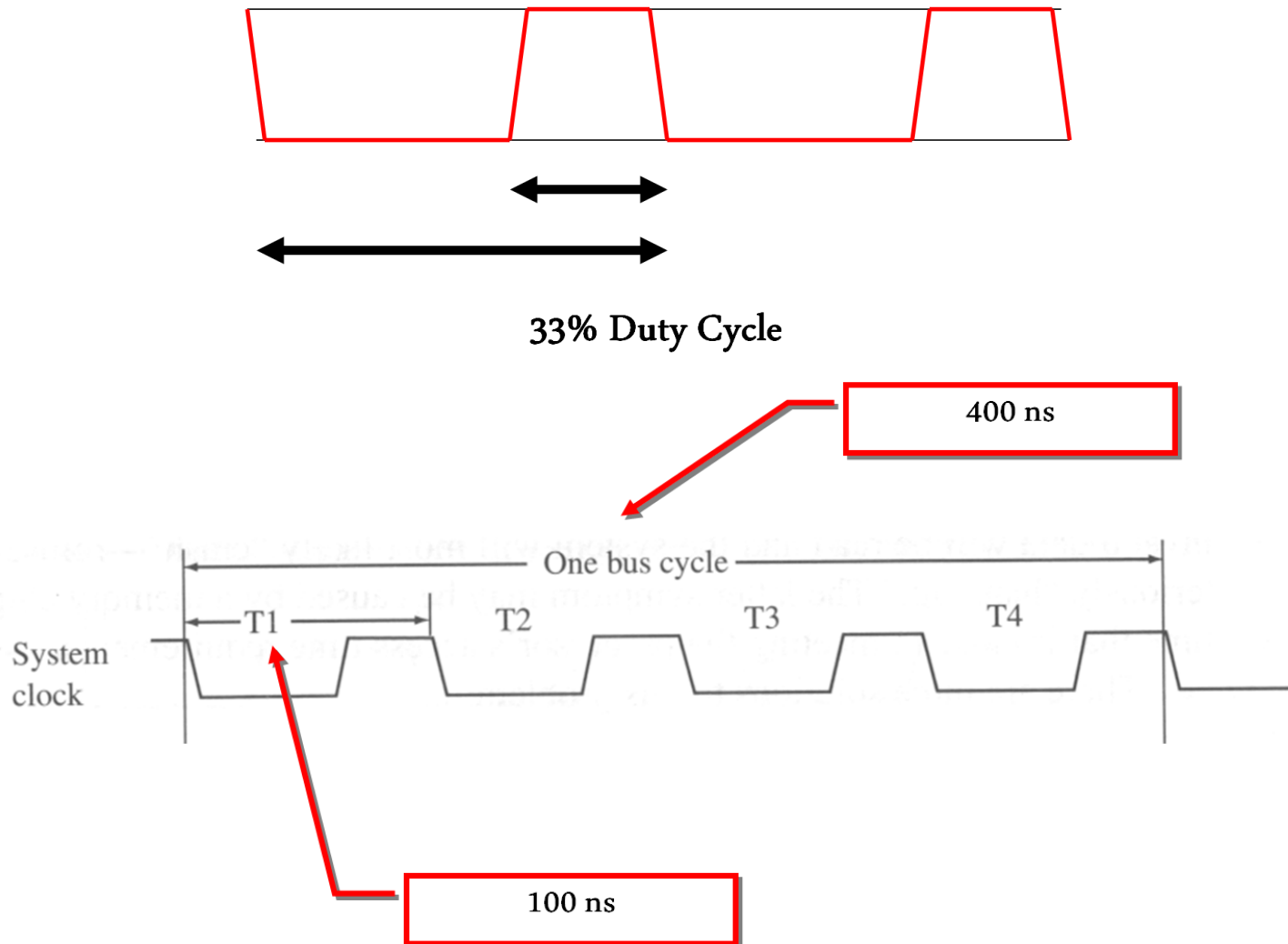
- ▶ The time a μ P requires to complete **fetch-decode-execute** operation of a single instruction is known as **Instruction Cycle**
- ▶ An **Instruction Cycle** consists of **one or more Machine Cycles**
- ▶ A basic μ P operation such as **reading or writing a byte/word from or to memory or I/O port** is called a **Machine Cycle or Bus cycle**
- ▶ A **Machine (bus) cycle** consists of at least **four clock cycles, called T states.**
- ▶ **One cycle of a clock is called a State**

Clock Generation

- ▶ Clock generator circuit is 8254A and connected to **pin 19 (CLK)** of 8086.



System Clock Concept



System Clock Concept

- ▶ 8086 is found to operate in between 5 to 10 Mhz.
- ▶ Each **bus cycle** consists of at least 4 clock cycles.
- ▶ An 8086 running at 5MHz, it's clock pulses will be of 200ns and it would take 800ns for a complete bus cycle.
- ▶ Again, an 8086 running at 10MHz, it's clock pulses will be of 100ns and it would take 400ns for a complete bus cycle.
- ▶ Each **read** or **write** operation take 1 bus cycles.

Clock States

► Why are there T states?

- In the 8086, the address and data lines are multiplexed.
- The microprocessor needs time to change the signals during each bus cycle.
- Memory devices need time to interpret the address value and then **read/write** the data (*access time*)

Clock States

- ▶ A specific, defined action occurs during each T states (labeled $T_1 - T_4$)
- ▶ **T_1 : Address is output**
 - ▶ Address of memory or I/O is sent out by 8086 via address bus
 - ▶ Used Control signals: ALE, DT/R', M/IO' shows some output
- ▶ **T_2 : Bus cycle type (MEMORY/IO, READ/WRITE)**
 - ▶ 8086 issues either RD' or WR' and DEN'
 - ▶ In case of **WRITE (WR)** operation, data to be written appear on data bus

Clock States

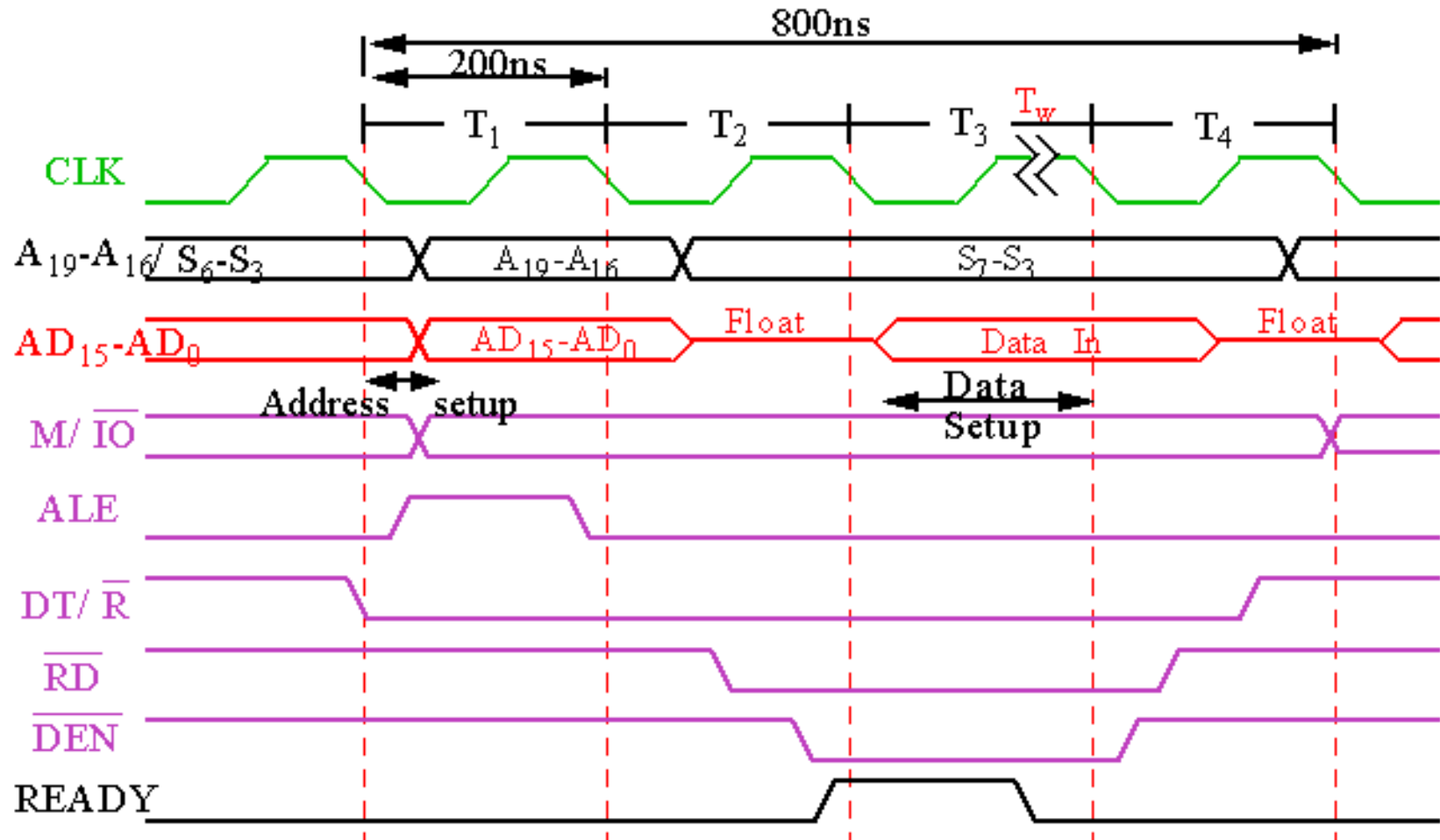
▶ ***T₃: Data is supplied***

- ▶ READY is sampled at the end of T2
 - ▶ If READY is low, T3 becomes a wait state (TW), means no operation (NOP).
 - ▶ In **READ** bus cycle data bus is sampled at end of T₃

▶ ***T4: Data latched by μ P, and control signals removed***

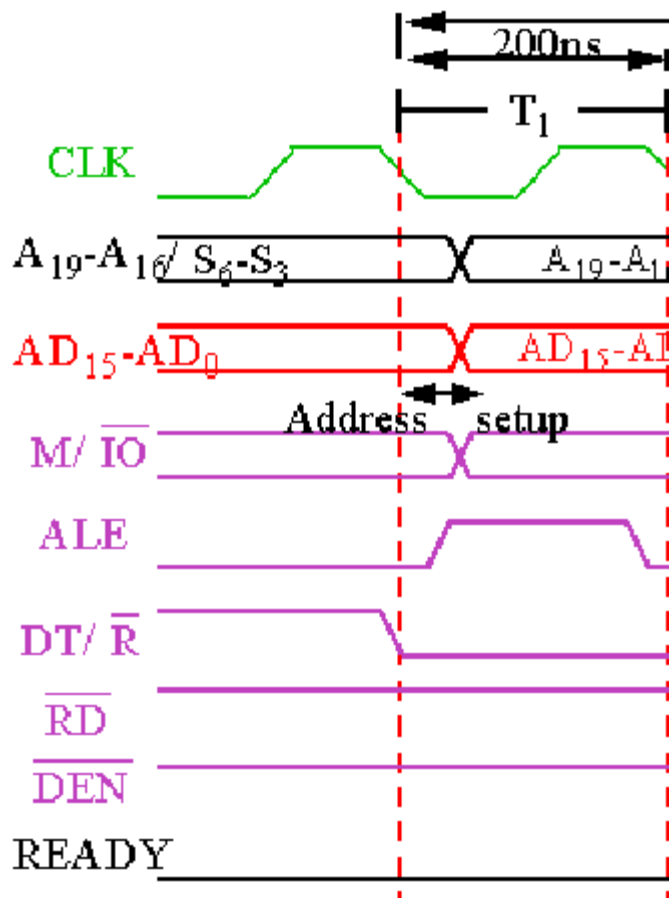
- ▶ All bus signals deactivated in preparation for next bus cycle
- ▶ μ P sampled data bus for data that read from M or I/O
- ▶ At trailing edge of VVR', transfer data to M or I/O

READ BUS Timing (Complete BUS Cycle)



Bus Timing for a Read Operation

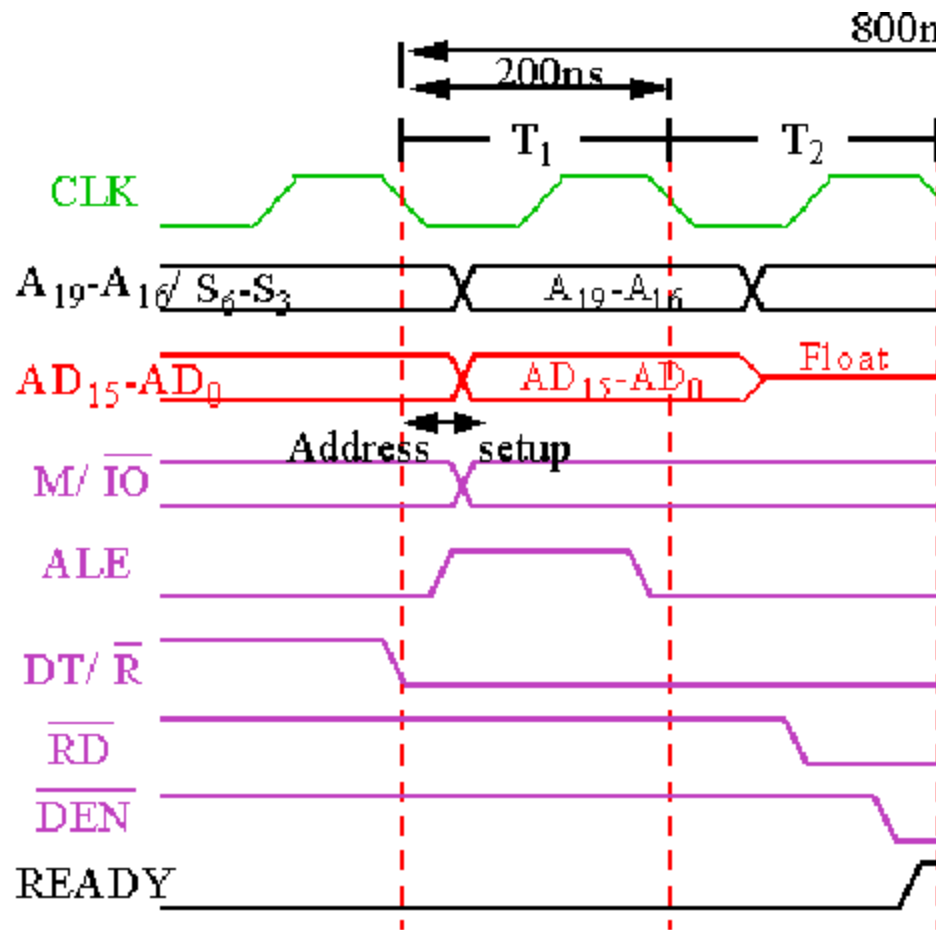
READ BUS Timing (During T_1 State)



During T_1 :

- The address is placed on the Address/Data bus.
- Control signals
 - **$M/\overline{IO}$** specify memory or I/O,
 - **ALE** latch the address onto the address bus and
 - **$DT/\overline{R}$** set the direction of data transfer on data bus.

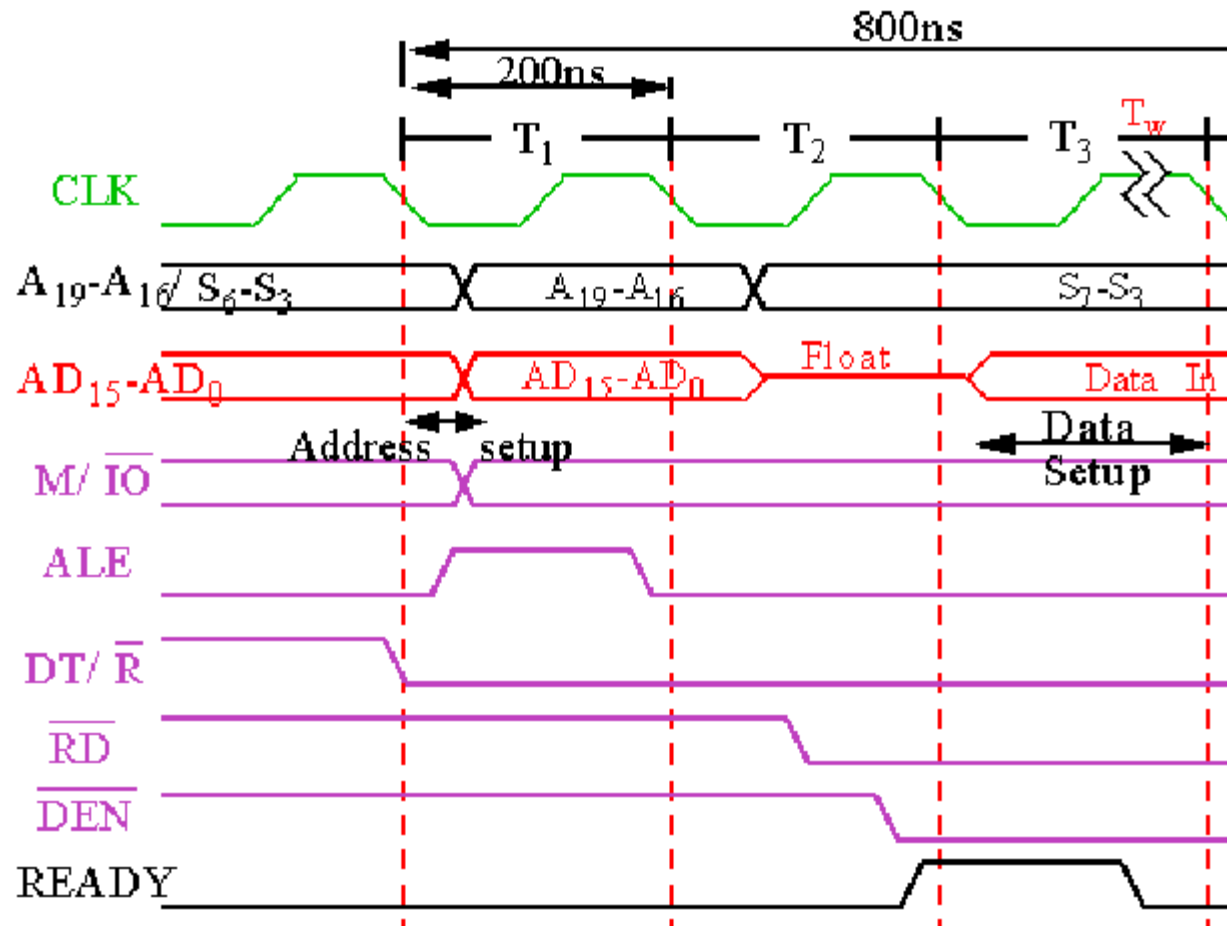
READ BUS Timing (During T_2 State)



During T_2 :

- 8086 issues the **RD'** (or **WR'** in case of write operation) signal.
- **DEN'** enables the 8086 to receive the data for **READ** operation (or the memory or I/O device to receive the data for **WRITE** operation).

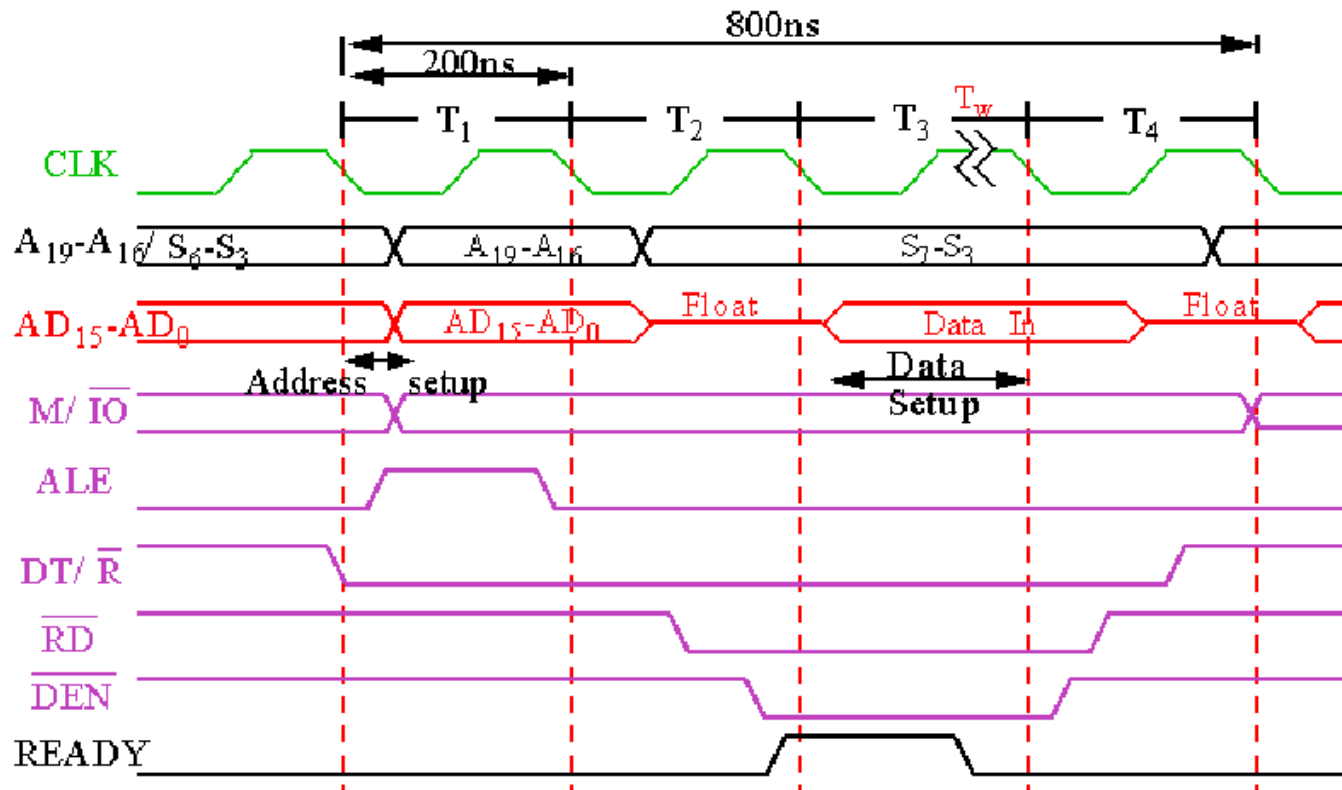
READ BUS Timing (During T_3 State)



During T_3 :

- This cycle is provided to allow memory to access data.
- **READY** is sampled at the end of T_2 .
 - If low, T_3 becomes a **wait** state.
 - Otherwise, the data bus is sampled at the end of T_3 .

READ BUS Timing (During T_4 State)

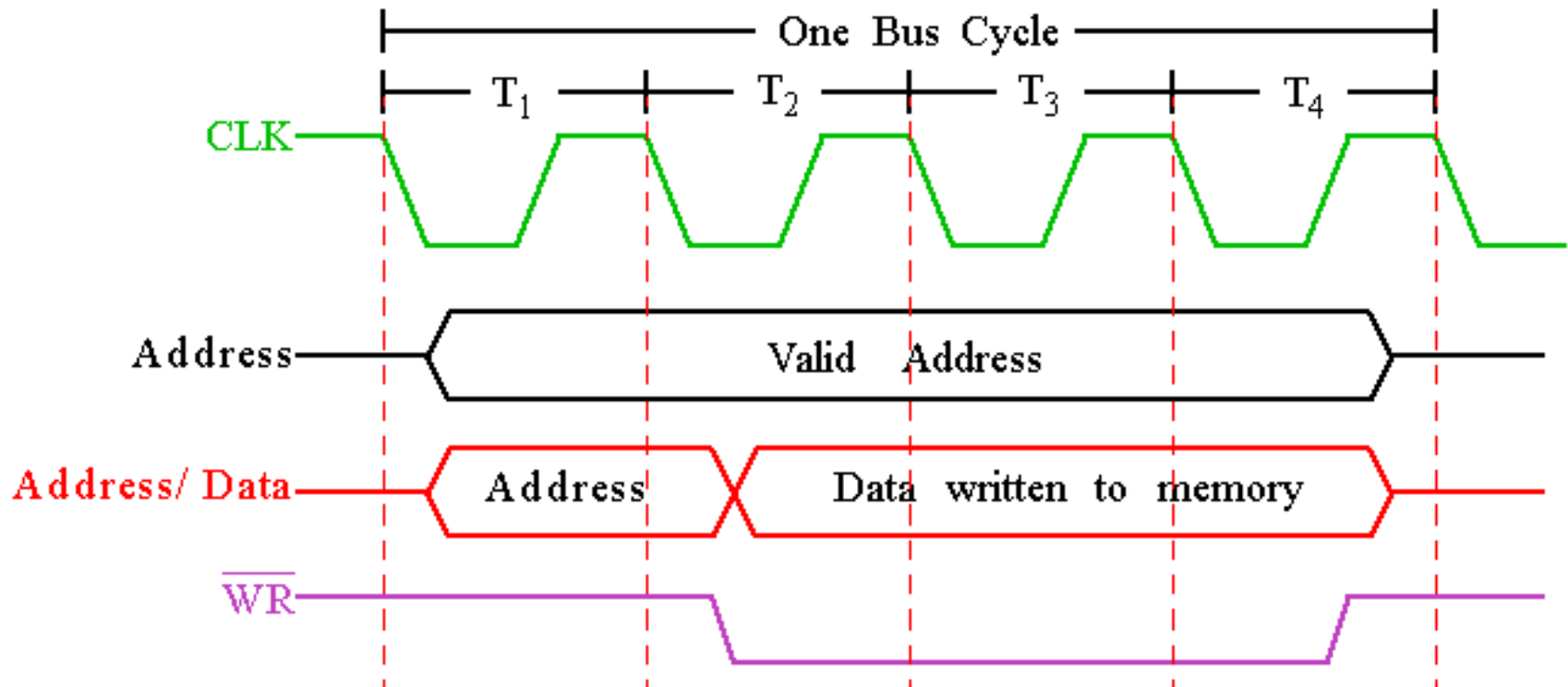


Bus Timing for a Read Operation

During T_4 :

- All bus signals are deactivated, in preparation for next bus cycle.
- Data is sampled for **READ** (or **WRITE** occurs for write) data.

WRITE BUS Timing



Simplified 8086 Write Bus Cycle

- ▶ What are the functions of each pin in different **T states** during **WRITE** operation ??

Thank You !!

